Inference for categorical data

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Getting Started

Load packages

In this lab, we will explore and visualize the data using the **tidyverse** suite of packages, and perform statistical inference using **infer**. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let's load the packages.

```
library(tidyverse)
library(openintro)
library(infer)
```

```
options(tibble.width = Inf)
```

The data

You will be analyzing the same dataset as in the previous lab, where you delved into a sample from the Youth Risk Behavior Surveillance System (YRBSS) survey, which uses data from high schoolers to help discover health patterns. The dataset is called `yrbss`.

1. What are the counts within each category for the amount of days these students have texted while driving within the past 30 days?

Insert your answer here

```
view(yrbss)
```

```
yrbss %>%
  count(text_while_driving_30d)
```

```
## # A tibble: 9 x 2
##   text_while_driving_30d     n
##   <chr>                  <int>
## 1 0                       4792
## 2 1-2                      925
## 3 10-19                    373
## 4 20-29                    298
## 5 3-5                      493
## 6 30                       827
## 7 6-9                      311
## 8 did not drive           4646
## 9 <NA>                     918
```

Among the 13,583 students: 0 days = 4,792, 1-2 days = 925, 3-5 days = 493, 6-9 days = 311, 10-19 days = 373, 20-29 days = 298, all 30 days = 827, did not drive = 4,646, missing = 918.

2. What is the proportion of people who have texted while driving every day in the past 30 days and never wear helmets?

Insert your answer here

```
yrbss %>%
  filter(helmet_12m == "never") %>%
  mutate(text_ind = ifelse(text_while_driving_30d == "30", "yes", "no")) %>%
  drop_na(text_ind) %>%
  count(text_ind) %>%
  mutate(prop = n / sum(n))
```

```
## # A tibble: 2 x 3
##   text_ind     n   prop
##   <chr>    <int>  <dbl>
## 1 no        6040 0.929
## 2 yes        463 0.0712
```

Among non-helmet wearers with a response, 463 of 6,503 texted while driving every day, so the proportion is about 0.071.

Remember that you can use `filter` to limit the dataset to just non-helmet wearers. Here, we will name the dataset `no_helmet`.

```
data('yrbss', package='openintro')
no_helmet <- yrbss %>%
  filter(helmet_12m == "never")
```

Also, it may be easier to calculate the proportion if you create a new variable that specifies whether the individual has texted every day while driving over the past 30 days or not. We will call this variable `text_ind`.

```
no_helmet <- no_helmet %>%
  mutate(text_ind = ifelse(text_while_driving_30d == "30", "yes", "no"))
```

Inference on proportions

When summarizing the YRBSS, the Centers for Disease Control and Prevention seeks insight into the population *parameters*. To do this, you can answer the question, "What proportion of people in your sample reported that they have texted while driving each day for the past 30 days?" with a statistic; while the question "What proportion of people on earth have texted while driving each day for the past 30 days?" is answered with an estimate of the parameter.

The inferential tools for estimating population proportion are analogous to those used for means in the last chapter: the confidence interval and the hypothesis test.

```
no_helmet %>%
  drop_na(text_ind) %>% # Drop missing values
  specify(response = text_ind, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1   0.0650   0.0781
```

Note that since the goal is to construct an interval estimate for a proportion, it's necessary to both include the `success` argument within `specify`, which accounts for the proportion of non-helmet wearers than have

consistently texted while driving the past 30 days, in this example, and that `stat` within `calculate` is here "prop", signaling that you are trying to do some sort of inference on a proportion.

3. What is the margin of error for the estimate of the proportion of non-helmet wearers that have texted while driving each day for the past 30 days based on this survey?

Insert your answer here

```
set.seed(607)
nohelmet_ci <- no_helmet %>%
  drop_na(text_ind) %>%
  specify(response = text_ind, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)

nohelmet_ci
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1   0.0650   0.0773
```

```
(nohelmet_ci$upper_ci - nohelmet_ci$lower_ci) / 2
```

```
## [1] 0.006151007
```

The 95% interval is roughly (0.065, 0.078), so the margin of error is about 0.006.

4. Using the `infer` package, calculate confidence intervals for two other categorical variables (you'll need to decide which level to call "success", and report the associated margins of error. Interpet the interval in context of the data. It may be helpful to create new data sets for each of the two countries first, and then use these data sets to construct the confidence intervals.

Insert your answer here

I chose the proportion of students who are Hispanic and the proportion who sleep fewer than 5 hours on school nights.

```
hispanic_ci <- yrbss %>%
  drop_na(hispanic) %>%
  specify(response = hispanic, success = "hispanic") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)

sleep5_ci <- yrbss %>%
  drop_na(school_night_hours_sleep) %>%
  mutate(sleep_lt5 = ifelse(school_night_hours_sleep == "<5", "yes", "no")) %>%
  specify(response = sleep_lt5, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)

bind_rows(hispanic = hispanic_ci, sleep_lt5 = sleep5_ci, .id = "variable") %>%
  mutate(margin_of_error = (upper_ci - lower_ci) / 2)
```

```
## # A tibble: 2 x 4
##   variable  lower_ci upper_ci margin_of_error
```

```
##   <chr>        <dbl>    <dbl>           <dbl>
## 1 hispanic    0.249    0.264         0.00772
## 2 sleep_lt5   0.0738   0.0831        0.00466
```

Hispanic (success = "hispanic"): $\hat{p} \approx 0.256$, 95% CI margin of error about 0.007. We are 95% confident that between about 24.9% and 26.4% of the high schoolers in the population this survey represents are Hispanic. Sleeps < 5 hours on school nights (success = "<5"): $\hat{p} \approx 0.078$, 95% CI margin of error about 0.005. We are 95% confident that between about 7.3% and 8.3% of these high schoolers sleep fewer than 5 hours on school nights.

The second interval is narrower because p is closer to 0, which gives a smaller $p(1-p)$.

How does the proportion affect the margin of error?

Imagine you've set out to survey 1000 people on two questions: are you at least 6-feet tall? and are you left-handed? Since both of these sample proportions were calculated from the same sample size, they should have the same margin of error, right? Wrong! While the margin of error does change with sample size, it is also affected by the proportion.

Think back to the formula for the standard error: $SE = \sqrt{p(1-p)/n}$. This is then used in the formula for the margin of error for a 95% confidence interval:

$$ME = 1.96 \times SE = 1.96 \times \sqrt{p(1-p)/n}\,.$$

Since the population proportion p is in this ME formula, it should make sense that the margin of error is in some way dependent on the population proportion. We can visualize this relationship by creating a plot of ME vs. p.

Since sample size is irrelevant to this discussion, let's just set it to some value ($n = 1000$) and use this value in the following calculations:

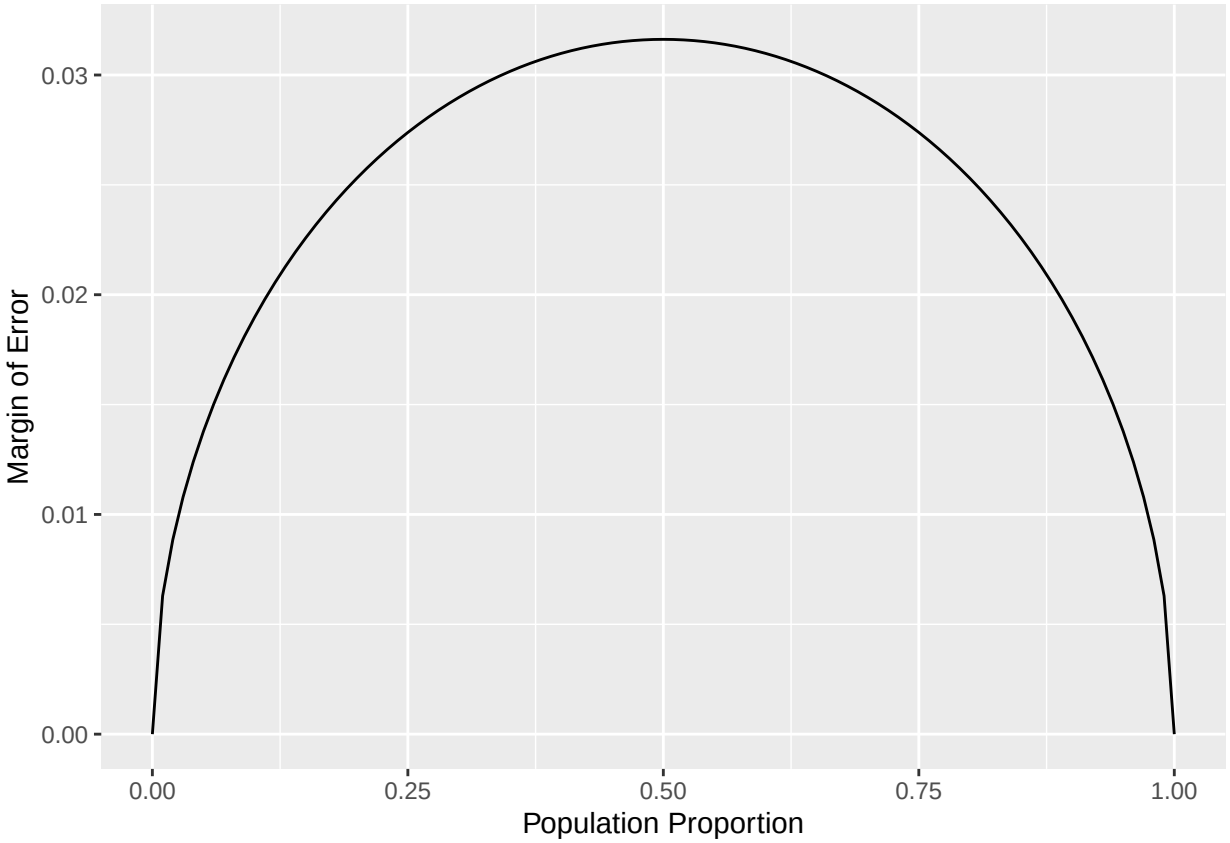
```
n <- 1000
```

The first step is to make a variable `p` that is a sequence from 0 to 1 with each number incremented by 0.01. You can then create a variable of the margin of error (`me`) associated with each of these values of `p` using the familiar approximate formula ($ME = 2 \times SE$).

```
p <- seq(from = 0, to = 1, by = 0.01)
me <- 2 * sqrt(p * (1 - p)/n)
```

Lastly, you can plot the two variables against each other to reveal their relationship. To do so, we need to first put these variables in a data frame that you can call in the `ggplot` function.

```
dd <- data.frame(p = p, me = me)
ggplot(data = dd, aes(x = p, y = me)) +
  geom_line() +
  labs(x = "Population Proportion", y = "Margin of Error")
```



5. Describe the relationship between `p` and `me`. Include the margin of error vs. population proportion plot you constructed in your answer. For a given sample size, for which value of `p` is margin of error maximized?

Insert your answer here

The relationship is a symmetric, inverted-U (parabola-like) curve. The margin of error is 0 at $p = 0$ and $p = 1$, rises as p moves toward the middle, and is largest at $p = 0.5$, where $p(1 - p)$ is maximized. So for a fixed sample size, proportions near 0.5 are estimated least precisely and proportions near 0 or 1 most precisely.

Success-failure condition

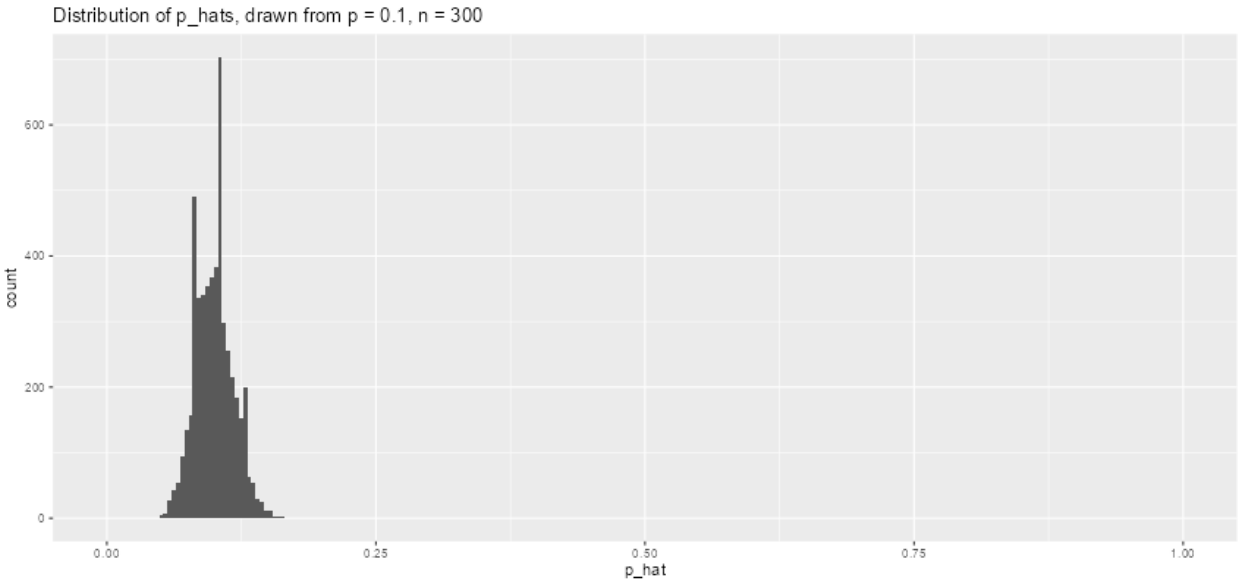
We have emphasized that you must always check conditions before making inference. For inference on proportions, the sample proportion can be assumed to be nearly normal if it is based upon a random sample of independent observations and if both $np \geq 10$ and $n(1 - p) \geq 10$. This rule of thumb is easy enough to follow, but it makes you wonder: what's so special about the number 10?

The short answer is: nothing. You could argue that you would be fine with 9 or that you really should be using 11. What is the "best" value for such a rule of thumb is, at least to some degree, arbitrary. However, when np and $n(1 - p)$ reaches 10 the sampling distribution is sufficiently normal to use confidence intervals and hypothesis tests that are based on that approximation.

You can investigate the interplay between n and p and the shape of the sampling distribution by using simulations. Play around with the following app to investigate how the shape, center, and spread of the distribution of $\hat{p}$ changes as n and p changes.

6. Describe the sampling distribution of sample proportions at $n = 300$ and $p = 0.1$. Be sure to note the center, spread, and shape.

Insert your answer here



The center is at around 0.10. The spread seems to be somewhere around 0.06 and 0.14. It looks like a unimodal shape.

7. Keep n constant and change p. How does the shape, center, and spread of the sampling distribution vary as p changes. You might want to adjust min and max for the x-axis for a better view of the distribution.

Insert your answer here

With n fixed at 300, the sampling distribution is always centered at about p, so it slides along the x-axis as p changes. The spread is largest at p = 0.5 and shrinks as p approaches 0 or 1. Near p = 0.5 the shape is symmetric and approximately normal. As p gets small, the distribution becomes right-skewed and is truncated at 0, and as p gets large it becomes left-skewed.

8. Now also change n. How does n appear to affect the distribution of $\hat{p}$?

Insert your answer here

The center stays at p, but increasing n makes the distribution narrower. A larger n also makes the shape more symmetric and normal, because np and $n(1-p)$ grow. A small n produces a wider, lumpier, skewed distribution, especially when p is near 0 or 1.

More Practice

For some of the exercises below, you will conduct inference comparing two proportions. In such cases, you have a response variable that is categorical, and an explanatory variable that is also categorical, and you are comparing the proportions of success of the response variable across the levels of the explanatory variable. This means that when using `infer`, you need to include both variables within `specify`.

9. Is there convincing evidence that those who sleep 10+ hours per day are more likely to strength train every day of the week? As always, write out the hypotheses for any tests you conduct and outline the status of the conditions for inference. If you find a significant difference, also quantify this difference with a confidence interval.

Insert your answer here

Hypotheses: Let p_{10+} be the proportion of students sleeping 10+ hours who strength train all 7 days, and $p_{<10}$ the proportion among students sleeping fewer than 10 hours.

- $H_0 : p_{10+} = p_{<10}$ sleep is unrelated to training every day
- $H_A : p_{10+} > p_{<10}$ 10+ hour sleepers are more likely to train every day

```
sleep_strength <- yrbss %>%
  drop_na(school_night_hours_sleep, strength_training_7d) %>%
  mutate(sleep_10 = ifelse(school_night_hours_sleep == "10+", "10+", "<10"),
         strength_ind = ifelse(strength_training_7d == 7, "yes", "no"))
```

```
sleep_strength %>%
  count(sleep_10, strength_ind)
```

```
## # A tibble: 4 x 3
##   sleep_10 strength_ind     n
##   <chr>    <chr>        <int>
## 1 10+      no             228
## 2 10+      yes             84
## 3 <10      no            9949
## 4 <10      yes           1958
```

```
obs_diff <- sleep_strength %>%
  specify(strength_ind ~ sleep_10, success = "yes") %>%
  calculate(stat = "diff in props", order = c("10+", "<10"))
```

```
set.seed(607)
null_dist <- sleep_strength %>%
  specify(strength_ind ~ sleep_10, success = "yes") %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in props", order = c("10+", "<10"))

obs_diff
```

```
## Response: strength_ind (factor)
## Explanatory: sleep_10 (factor)
## # A tibble: 1 x 1
##    stat
##   <dbl>
## 1 0.105
```

```
null_dist %>% get_p_value(obs_stat = obs_diff, direction = "greater")
```

```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1       0
```

Conditions: The YRBSS is a random sample well under 10% of all U.S. high schoolers, and the two groups are separate students, so independence holds. With a pooled proportion of about 0.167, all expected counts (about 52, 260, 1,990 and 9,917) are well above 10, so the success-failure condition holds.

Result: 26.9% of students sleeping 10+ hours, 84 of 312, strength train all 7 days, versus 16.4% of those sleeping less, 1,958 of 11,907, a difference of about 0.105. No permuted difference came close, so $p < 0.001$. We reject H0: there is convincing evidence that students who sleep 10+ hours are more likely to strength train every day.

```r
set.seed(607)
sleep_strength %>%
  specify(strength_ind ~ sleep_10, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in props", order = c("10+", "<10")) %>%
  get_ci(level = 0.95)
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1   0.0516    0.157
```

We are 95% confident that the proportion who strength train every day is about 5.5 to 15.4 percentage points higher among students who sleep 10+ hours than among those who sleep less.

10. Let's say there has been no difference in likeliness to strength train every day of the week for those who sleep 10+ hours. What is the probablity that you could detect a change (at a significance level of 0.05) simply by chance? *Hint:* Review the definition of the Type 1 error.

Insert your answer here

If there were truly no difference, the probability of detecting one anyway is 0.05 (5%). That is the Type I error rate, the probability of rejecting a true null hypothesis. About 1 in 20 such studies would show a difference purely by chance.

11. Suppose you're hired by the local government to estimate the proportion of residents that attend a religious service on a weekly basis. According to the guidelines, the estimate must have a margin of error no greater than 1% with 95% confidence. You have no idea what to expect for p. How many people would you have to sample to ensure that you are within the guidelines?
Hint: Refer to your plot of the relationship between p and margin of error. This question does not require using a dataset.

Insert your answer here

With no idea what p is, use the worst case $p = 0.5$, where the margin of error is largest. We need $1.96\sqrt{0.5(0.5)/n} \leq 0.01$:

$$n \geq \frac{1.96^2(0.5)(0.5)}{0.01^2} = 9603.04$$

We need to sample at least 9,604 people. Using this n, the margin of error is at most 1% for any true p.
